

STA304 Proposal: Workload and AI Use in Lecture and Tutorial Attendance

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1. Project Title

Weekly Workload and AI-Substituted Absence from Lectures and Tutorials among STA304H5 Students

2. Research Questions

This study examines whether students with heavier weekly workloads skip lectures and tutorials in favour of AI tools. It concerns study habits, not academic integrity.

- **RQ1:** Is heavier weekly coursework workload associated with skipping a larger percentage of lectures and tutorials?
- **RQ2:** Among students who have skipped both lectures and tutorials, does the percentage of skipped classes attributed to AI differ between lectures and tutorials?
- **RQ3:** Among students who have skipped at least one type of class, does weekly coursework workload (hours per week) remain a significant predictor of the percentage of their skipped classes attributed to AI, after controlling for number of courses, paid work hours, program of study, and perceived AI substitutability?

3. Target Population

All students enrolled in STA304H5F (Surveys, Sampling and Observational Data) at the University of Toronto Mississauga in the Fall 2026 term. Participation is restricted exclusively to these students, and each respondent is identified only through their course-given ID.

4. Sampling Method

We will use simple random sampling (SRS). The sampling frame is the STA304H5F class list of $N = [N]$ students. Each student will be numbered 1 to N , and we will randomly select $n = [n]$ students without replacement using R (`sample(1:N, n, replace = FALSE)`), so every possible sample of size n has the same chance of being chosen. We chose SRS because our population is a single course whose students are fairly similar, and because it is the simplest probability

sampling design. Program of study is asked in the questionnaire and used as a predictor in RQ3, so we do not need to stratify on it.

5. Data Collection

Data collection will not begin until we receive feedback on this proposal. We will then post an anonymous online questionnaire (Google Form) on the course Piazza master list for a one-week window, with one reminder post halfway through. Students complete it on a volunteer basis, so Section 4 describes the sampling design we would use with full control over recruitment, and we will note volunteer participation as a limitation when interpreting results.

To secure an adequate sample size, we offer a reciprocal incentive: any student who completes our questionnaire can reply with their own group's link, and we will complete their questionnaire within 24 hours. This complements the course requirement to complete questionnaires for marks. We also keep the survey short (about 3 minutes) and post one reminder halfway through the week.

Participation is voluntary and anonymous. Some students may not respond, so results describe respondents and may not fully represent all STA304H5 students.

6. Statistical Tests

Workload, number of courses, and paid work hours are coded by category order (workload and courses 1 to 5, paid work hours 1 to 4). Skipping percentage uses each bin's midpoint (0, 13, 38, 63, 88), averaged across lectures and tutorials (lectures only if a student has no tutorials). The AI-skipping percentage (Q7, Q8: the percentage of skipped classes attributed to AI) uses the same midpoints, and perceived AI substitutability is coded 1 (Strongly disagree) to 5 (Strongly agree). "I have not skipped any", "None of my courses have tutorials", and "Prefer not to disclose" are treated as missing, while "0%" is coded as 0.

Simple tests

- **Correlation and simple linear regression (RQ1).** Workload is the predictor and skipping percentage is the response. The correlation and regression slope show the direction and strength of the association.
- **Paired t-test (RQ2).** Each student's lecture AI-skipping percentage is compared with their tutorial percentage. Only students who skipped both can be included, so this sample will be smaller.

Advanced test

- **Multiple linear regression with model selection (RQ3).** The response is the AI-skipping percentage (the average of the lecture and tutorial percentages, using whichever is available). The predictors are workload, number of courses, paid work

hours, program of study, and perceived AI substitutability (6 coefficients, counting two indicator variables for program of study). Workload stays in every candidate model because it is the predictor of interest, and we compare models with different sets of the other predictors using adjusted R^2 and residual plots. We will report workload's coefficient, 95% confidence interval, and p-value from the selected model.

Sample size. Six coefficients need about 60 usable responses (roughly 10 per coefficient). Some students will drop out (never skipped, or missing answers), so we aim for at least [100] completed responses. All tests use a 5% significance level.

7. Questionnaire (not counted in the 2-page limit)

Survey link:

https://docs.google.com/forms/d/e/1FAIpQLSdnCYBckBLNBDC_p98w3L4mVduz5-bKDsbbiV6cHzczznxnUQ/viewform?usp=publish-editor

Introduction shown to participants

Welcome, and thanks for helping with our STA304 project!

We are a group of STA304H5 students studying how weekly workload relates to class attendance and the use of AI tools for studying. The survey has 9 short questions plus your course ID and takes about 3 minutes. A few things to know before you start:

- Taking part is your choice. You can close the form at any time without submitting.
- Your answers are anonymous. Your course specific ID is a code, not your name, and it is used only to confirm that you completed the survey for course credit.
- Answers are used only for this course project and are reported as group summaries.
- We are asking about study habits, not academic integrity, so please answer honestly.

By submitting this form, you agree to take part. Questions? Contact [group member name, UofT email]. Group [number].

Q0. What is your assigned course specific ID? (short answer, required)

Q1. In a typical week this term, about how many hours do you spend on coursework across all of your courses? Include assignments, studying, and projects, and exclude time in scheduled lectures and tutorials.

- Less than 10 hours
- 10 to 19 hours
- 20 to 29 hours
- 30 to 39 hours
- 40 hours or more

Q2. How many courses are you taking this term?

- 1 to 2
- 3
- 4
- 5
- 6 or more

Q3. In a typical week this term, about how many hours do you spend in paid work?

- None
- 1 to 9 hours
- 10 to 19 hours
- 20 hours or more
- Prefer not to disclose

Q4. Which best describes your program of study?

- Statistics
- Mathematics
- Computer Science
- Other

Q5. So far this term, about what percentage of your scheduled lectures (across all your courses) have you skipped?

- 0%
- 1 to 25%
- 26 to 50%
- 51 to 75%
- 76 to 100%

Q6. So far this term, about what percentage of your scheduled tutorials (across all your courses) have you skipped?

- 0%
- 1 to 25%
- 26 to 50%
- 51 to 75%
- 76 to 100%
- None of my courses have tutorials

Q7. Of the lectures you have skipped this term, about what percentage did you skip at least partly because you planned to use AI tools to cover the material?

- I have not skipped any lectures

- 0% (none of the lectures I skipped)
- 1 to 25%
- 26 to 50%
- 51 to 75%
- 76 to 100%

Q8. Of the tutorials you have skipped this term, about what percentage did you skip at least partly because you planned to use AI tools to cover the material?

- I have not skipped any tutorials
- None of my courses have tutorials
- 0% (none of the tutorials I skipped)
- 1 to 25%
- 26 to 50%
- 51 to 75%
- 76 to 100%

Q9. How much do you agree with this statement: "Studying with AI tools is an adequate replacement for attending class in person."

- Strongly agree
- Agree
- Neither agree nor disagree
- Disagree
- Strongly disagree